Emily L. Hunt – Curriculum Vitae

On request | emily.hunt.physics@gmail.com | emily.space | emilyhunt

Research Profile

Astronomer with interests in machine learning and statistics. Highly skilled programmer with 10+ years of programming experience. During my Ph.D., I used Gaia data and various machine learning techniques to create the largest ever catalogue of star clusters in the Milky Way. I am looking to work on applications of machine learning to large astronomical datasets such as Gaia, Vera Rubin, and JWST surveys.

Education & Employment

2025-29, Postdoc, University of Vienna, Austria

2025-25, Postdoc, Max Planck Institute for Astronomy, Germany

2023-24, Postdoc, Heidelberg University, Germany

Ph.D. 2023, Heidelberg University, Germany
Thesis: "Improving the census of open clusters in the Milky Way with data from Gaia"
Advisor: S. Reffert

M.Phys. 2019, University of Bath, United Kingdom
Thesis: "Inference of photometric galaxy redshifts with a mixture density network"
Advisor: S. Wuyts

Publications

ADS search

First author

6. **Emily L. Hunt**, Vincent S. Carpenter, Kyle W. Cook *et al.* (2026). "The Astrosky Ecosystem: An independent online platform for science communication and social networking". ADASS XXXV (Conference proceedings)

5. **Emily L. Hunt**, Tristan Cantat-Gaudin, Friedrich Anders *et al.* (2026). "The selection function of the Gaia DR3 open cluster census". A&A, 706, A341

4. **Emily L. Hunt**, Tristan Cantat-Gaudin, Friedrich Anders *et al.* (2025). "The completeness of the open cluster census towards the Galactic anticentre". A&A, 699, A273

3. **Emily L. Hunt** and Sabine Reffert (2024). "Improving the open cluster census. III. Using cluster masses, radii, and dynamics to create a cleaned open cluster catalogue". A&A, 686, A42
(175 citations)

2. **Emily L. Hunt** and Sabine Reffert (2023). "Improving the open cluster census. II. An all-sky cluster catalogue with Gaia DR3". A&A, 673, A114
(336 citations)

1. **Emily L. Hunt** and Sabine Reffert (2021). "Improving the open cluster census. I. Comparison of clustering algorithms applied to Gaia DR2 data". A&A, 646, A104
(129 citations)

Co-author

6. Kristen C. Dage, **Emily L. Hunt** *et al.* (2026). "The Secret Lives of Open Clusters: a Multiwavelength Examination of Three Open Clusters". PASA, accepted

5. Alexis L. Quintana, **Emily L. Hunt**, and Hanna Parul (2025). "How many stars form in compact clusters in the local Milky Way?". A&A, 701, L2

4. Richard I. Anderson and **Emily L. Hunt** (2025). "A birds-eye view of stellar evolution through populations of variable stars in Galactic open clusters". A&A, 700, L13

3. Sebastian Ratzenböck, João Alves, **Emily L. Hunt** *et al.* (2025). "Toward the fabric of the Milky Way: I. The density of disk streams from a local 250^3 pc^3 volume". A&A, 694, A307

2. Dane Spaeth, Sabine Reffert, **Emily L. Hunt** *et al.* (2024). "Non-radial oscillations mimicking a brown dwarf orbiting the cluster giant NGC 4349 No. 127". A&A, 689, A91

1. Cameren Swiggum *et al.* (incl. **Emily L. Hunt**) (2024). "Most nearby young star clusters formed in three massive complexes ". Nature, 661, 8019, p.49-53
(25 citations)

Selected Presentations

Invited talk, Astro-COLIBRI workshop 5 – Paris, France	(upcoming) 2026
Talk, Star Formation Fields – Rethymno, Greece	(upcoming) 2026
Invited talk, EAS (S11) – Lausanne, Switzerland	2026
Talk, EAS (S7) – Lausanne, Switzerland	2026
Invited talk, ATScience 2026 – Vancouver, Canada	2026
Seminar, TASTY – University of Toronto, Canada	2026
Seminar – Federal University of Rio Grande do Sul, Brazil	2026
Talk, Wienerwald Astronomy Symposium – ISTA, Austria	2025
Seminar, Department of Astrophysics – Lund University, Sweden	2025
Talk, ADASS XXXV – Görlitz, Germany	2025
Seminar, Virtual Astronomy Software Talks (VAST) – Online	2025
Talk, Bridging Scales – Matera, Italy	2025
Invited talk, EAS (SS45) – Cork, Ireland	2025
Talk, EAS (S9) – Cork, Ireland	2025
Talk, EAS (S1) – Cork, Ireland	2025

Colloquium – ARI, Heidelberg, Germany 2025
Talk, Heidelberg-Harvard Star Formation Workshop – Heidelberg, Germany 2024
Seminar, Stars seminar – Geneva, Switzerland 2024
Talk, MW Methods Workshop – Ringberg, Germany 2024
Invited talk, EAS (SS33) – Padova, Italy 2024
Talk, EAS (S4) – Padova, Italy 2024
Invited talk, SFML2024 – Budapest, Hungary 2024
Colloquium – University of Vienna, Austria 2024
Talk, From star clusters to field populations – Florence, Italy 2023
Seminar, CEFCA – Teruel, Spain (online) 2023
Talk, .Astronomy 12 – Flatiron Institute, New York, NY, USA 2023
Colloquium, Königstuhl Colloquium – MPIA, Heidelberg, Germany 2023
Talk, National Astronomy Meeting – Coventry, England, UK 2022
Invited talk, EAS (SS34) – Valencia, Spain 2022
Talk, EAS (SS24) – Valencia, Spain 2022
Talk, EAS (SS15) – Valencia, Spain 2022
Talk, LGBTQ+ STEMinar – University of Glasgow, Scotland, UK 2022
Seminar, Galaxy group – ARI, Heidelberg, Germany 2021
Seminar, Astronomy group – University of Hertfordshire, England, UK 2021
Talk, Star Clusters: The Gaia Revolution 2021
Invited talk, EAS (SS32) – Leiden, Netherlands 2021
Talk, EAS (S15) – Leiden, Netherlands 2021
Seminar, SFB 881 – Heidelberg, Germany 2021
Seminar, Gaia group – University of Vienna, Austria 2021
Seminar, Astronomy group – University of Bath, England, UK 2020
Seminar, Milky Way group – MPIA, Heidelberg, Germany 2020

Open-source software

The Astrosky Ecosystem – founder & lead developer of The Astrosky Ecosystem, which is building astronomy outreach and communication tools for Bluesky/AT Protocol applications that are used by over 10,000 people a day

ocelot – lead developer of an upcoming star cluster analysis Python package

The Star Formation Newsletter – built a new, arXiv API-linked website for the Star Formation Newsletter's 2026 relaunch

Teaching

Python Programming for Scientists, University of Vienna 2026
ISM & Star Formation*, University of Vienna 2025
Machine learning*, MWGaia Dr. Schl., University of Coimbra, Portugal 2024
Astronomy Lab Course, Heidelberg University 2021

Introduction to Astronomy I, Heidelberg University 2020

* = as a primary lecturer

Supervision

MSc supervisor: Janine Jochum 2026-27
BSc supervisor: 2 total 2026-
PhD committee: Kayvon Sharifi (Atlanta, GA, US) 2026-

Awards

Ernst Patzer Award for an excellent publication (press release) €2000 – 2023
University of Bath IMI Undergraduate Research Internship £2000 – 2018

Selected Outreach

Invited talk – OUTer SPACE, Max Planck Institute for Astronomy 2023
Interviewed for article – Space.com 2021
Interviewed for article – Thrillist.com 2020
Radio interview – Deutschlandfunk (public radio) & Neue Zürcher Zeitung 2020

Meeting organization & service

Technical Editor for the Star Formation Newsletter 2026-
Co-Chair at EAS 2025: Symposium S3 (Cork, Ireland) 2025
SOC for Roman Galactic Plane Survey Workshop (online) 2025
SOC for .Astronomy 13 (Madrid, Spain) 2024
SOC for .Astronomy 12 (New York, NY, USA) 2023
Reviewer for A&A, ApJ, AJ, MNRAS ongoing

Workshops Attended

.Astronomy 13 – ESAC, Madrid, Spain 2024
.Astronomy 12 – Flatiron institute, New York, NY, USA 2023
CZS school on Scientific Machine Learning – Heidelberg, Germany 2023
GaiaUnlimited Community Workshop – Heidelberg, Germany 2022
..Astronomy – online 2020

Relevant expertise

Programming languages

Python: expert
JavaScript: advanced
C/C++: intermediate
Java: basic

Tools and scripting languages

Git/GitHub: expert
LaTeX: expert
ADQL/SQL: expert
HTML/CSS: intermediate

Languages

English: native speaker
German: intermediate